

IoT Sensing Systems

Instructor

Utku Günay Acer

Email: utku.acer@sabanciuniv.edu

Course Description

This course gives an in-depth study of the technologies involved in the Internet of Things with a specific focus on networked sensing systems. This includes processing sensor data and making inferences using it in various platforms at the edge of the network. This course has a strong emphasis on hands-on experience.

Prerequisites:

CS 204

A basic understanding of computer organization, operating systems, and networks is necessary. Ability to write computer programs in a high-level language (such as Python and C or C++). Familiarity with Linux and shell environments.

Covered Topics

1. **Introduction to Internet of Things:** Examples of mobile and embedded systems and sensors.
2. **Sensing:** Review of signal processing and machine learning on sensor data
3. **System Considerations:** Networking issues at the edge, energy preservation, local/remote computation.
4. **Privacy in Sensing:** Sensor data and privacy
5. **Mobile/Embedded Computing & Wearables:** On-device sensing with smartphones, embedded devices, and wearables
6. **Edge Computing:** Remote inference with sensor data
7. **Cool Applications:** Localization, activity tracking, smart homes, mobile health, etc.

Textbooks

No textbook is required.

Grading

Students will be assessed based on homework assignments, a presentation and a final project. They are encouraged to propose their projects. While assessments can be performed individually, they will form groups of 4 for the final project. There will be no exams